

# Data Management Plan

## GitHub Repository

<https://github.com/yashs14282001/Group-1-Team-Project.git>

### 1. Overview of the Datasets

This project utilises publicly available cricket player data from the official Board of Control for Cricket in India (BCCI) website. Datasets were generated through web scraping with Python libraries including Requests, BeautifulSoup, and Pandas. These datasets support data analysis and machine learning tasks focused on Indian international cricket players, including batting and bowling performance across Test, ODI, and T20I formats.

The project uses at least two datasets:

#### Dataset 1: Player Profile Dataset

This study utilizes two datasets obtained from the official BCCI website via web scraping. The Player Profile Dataset comprises general player information, including player name, grade category, playing role, batting style, bowling style, and profile URL. The Player Performance Statistics Dataset contains batting and bowling statistics across Test, ODI, and T20I formats, such as runs, strike rates, centuries, wickets, and economy rates. Data collection was conducted using Python libraries including Requests, BeautifulSoup, and Pandas, and the datasets were stored in CSV format for subsequent analysis and research.

#### Dataset 2: IPL Batting & Fielding Statistics Dataset

The IPL Batting and Fielding Statistics Dataset comprises season-wise batting and fielding performance data of IPL players, sourced from official cricket statistics pages. Key variables include matches played, runs scored, batting average, strike rate, centuries, half-centuries, boundaries, sixes, and catches. Data collection was conducted using web scraping techniques, and the dataset is stored in CSV format to facilitate analysis and research. This resource enables the examination of player performance trends across multiple IPL seasons.

### 2. Collection of Data

The data was collected through web scraping from the official BCCI website:

Official Website:

<https://www.bcci.tv/>

Primary Source Pages:

- <https://www.bcci.tv/international/men/players>
- Individual player profile pages

Data collection was automated using Python scripts.

Libraries used:

- requests
- BeautifulSoup4
- pandas
- time

The scraper accessed public web pages to extract structured player information and cricket statistics.

The scraping workflow included:

1. Accessing the player listing page
2. Extracting profile URLs
3. Collecting batting and bowling statistics
4. Exporting results into CSV format

### 3. Metadata

#### Dataset Name

bcci\_players\_full\_stats.csv

#### Dataset Format

CSV (Comma Separated Values)

#### Data Type

Structured tabular data

#### Data Source

Official BCCI website : <https://www.bcci.tv/international/men/players>

#### Key Variables

The dataset includes key variables such as player name, grade category, playing role, batting and bowling styles, runs scored in Test, ODI, and T20I matches, strike rates, centuries, half-centuries, wickets taken, and economy rates. These variables support player performance analysis, cross-format statistical comparisons, and data analysis or machine learning applications.

#### File Encoding

UTF-8

#### Update Frequency

A static snapshot was collected during the project timeline.

### 4. Document Control

Version control and project collaboration are managed through GitHub.

GitHub is used for:

- Source code management
- Dataset storage
- Version tracking
- Team collaboration
- Documentation updates

#### File Naming Convention

The project uses a consistent file naming system:

| File Name                   | Description               |
|-----------------------------|---------------------------|
| bcci_scraper.py             | Main scraping script      |
| bcci_players_full_stats.csv | Final processed dataset   |
| <a href="#">README.md</a>   | Project documentation     |
| data_cleaning.py            | Data preprocessing script |
| analysis.ipynb              | Data analysis notebook    |

### 5. README File

The GitHub repository includes a [README.md](#) file with:

- Project overview
- Objectives
- Dataset description
- Setup instructions
- Required Python libraries
- Instructions to run the scraper
- Ethical considerations
- Team member contributions

The README supports project reproducibility and usability.

## 6. Security and Storage

Datasets and project files are securely stored using:

- GitHub private/public repository
- Local encrypted university systems
- Cloud backup storage

Security measures include:

- Restricted write access to repository collaborators
- Regular backups
- Version control for recovery
- No sensitive personal data stored

## 7. Ethical Statement

### 1. Does the data come under GDPR requirements?

Yes. The dataset mainly contains publicly available professional cricket statistics and player information published on the official BCCI website. No sensitive personal data, private records, or identifiable user information was collected. Therefore, the project does significantly fall under GDPR-sensitive processing requirements.

### 2. Does the project conform to UH ethical policies?

The project complies with the University of Hertfordshire's ethical policies by using only publicly accessible data and following transparent, non-intrusive data collection methods. The data is used strictly for academic research purposes, and the project does not involve human participants, surveys, or personal behavioral tracking.

### 3. Do you have permission to use the data for your proposed research project?

The data was collected from publicly accessible pages on the official BCCI website intended for public viewing. The project uses this data solely for educational and research purposes. The scraping process follows responsible usage practices and does not attempt to bypass security protections.

### 4. Are you assured that the data was collected in an ethical manner?

The original data is officially published and maintained by the Board of Control for Cricket in India (BCCI) through publicly accessible channels. This Data Management Plan ensures responsible data collection, secure storage, ethical usage, and proper documentation in line with university academic standards.